1.List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

1. Devices that use SMPS (Switched Mode Power Supply)

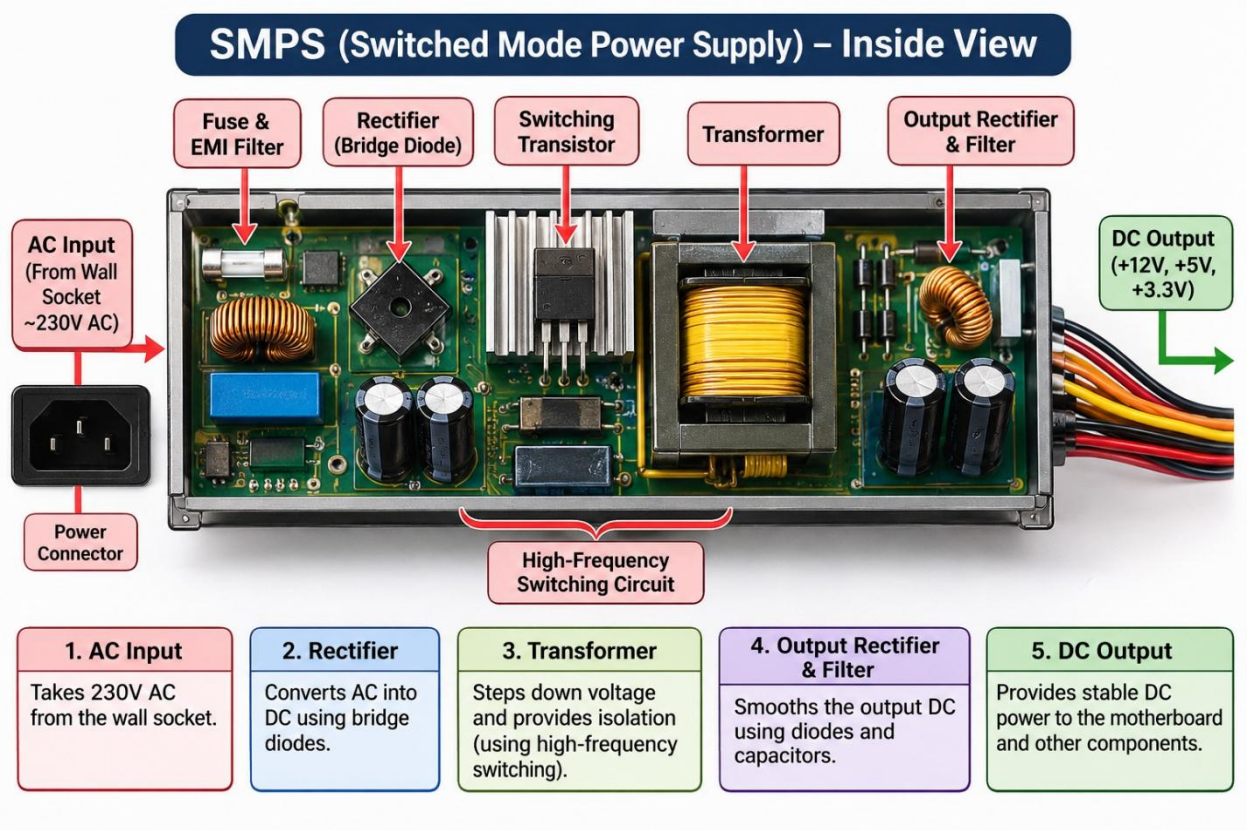
No.	Device	Why does it need SMPS?
1	Desktop Computer	Converts electricity into the correct voltage needed by the CPU, motherboard, and other components.
2	Laptop Charger	Converts high-voltage AC electricity into low-voltage DC power to charge the laptop.
3	LED Television	Converts electricity into the required power for the screen and internal circuits.

2. Devices that use UPS (Uninterruptible Power Supply)

No.	Device	Why does it need UPS?
1	Desktop Computer	Provides backup power during a power cut so we can save our work and shut down safely.
2	Wi-Fi Router	Keeps the internet connection working during short power cuts.

**2.Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.**



3.Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

1. ATX 24-pin motherboard connector



Detail	Explanation
Name	ATX 24-pin motherboard connector
Devices powered	Supplies the main power to the motherboard and its connected components.
Safety tip	Align the connector properly with the motherboard socket and press until the locking clip clicks. Never force it.

2. SATA Power Connector



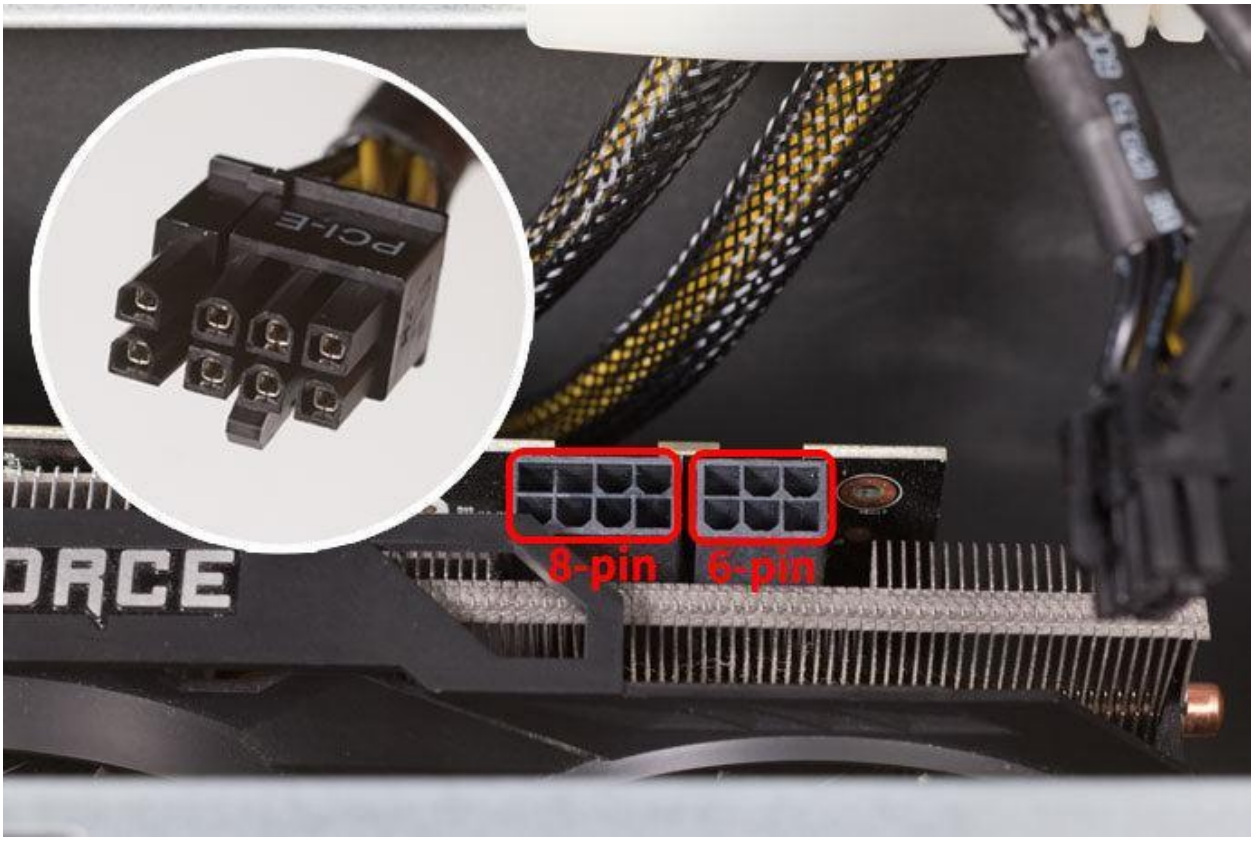
Detail	Explanation
Name	SATA power connector
Devices powered	Supplies power to SATA HDDs, SSDs, and DVD drives.
Safety tip	Make sure the L-shaped connector is aligned correctly before connecting it. Never force it into the socket.

3. Molex 4-pin Connector



Detail	Explanation
Name	Molex 4-pin power connector
Devices powered	Used for older hard drives, optical drives, and some cabinet fans.
Safety tip	Hold the connector firmly while removing it. Do not pull it by the wires.

4. PCIe 6-pin / 8-pin Connector



Detail	Explanation
Name	PCIe 6-pin / 8-pin power connector
Devices powered	Supplies additional power to graphics cards (GPU).
Safety tip	Use the correct PCIe/VGA power cable and make sure the locking clip is secured properly. Never confuse it with the CPU power connector.

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Step	Component	Cable / Connector Used	Description

1	Motherboard	24-pin ATX power connector	Connect the 24-pin cable from the SMPS to the motherboard to provide its main power supply.
2	CPU (Processor)	4-pin or 8-pin EPS/CPU power connector	Connect the CPU power cable from the SMPS to the motherboard near the processor.
3	Graphics Card (GPU)	PCIe 6-pin, 8-pin or 6+2-pin connector	Connect the PCIe power cable to the graphics card to provide the additional power needed for gaming.
4	HDD	SATA power connector	Connect the SATA power cable from the SMPS to the hard drive to supply electricity.
5	SATA SSD	SATA power connector	Connect the SATA power cable from the SMPS to the SSD to provide power.
6	Cabinet Fans (if required)	SATA or 4-pin Molex power connector	Connect the appropriate power cable to fans that require direct PSU power.